Hybrid Information Retrieval on Amazon Grocery Reviews: Combining BM25, Dense Retrieval, and RAG for Consumer Product Search

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Abstract

Effective product search in e-commerce requires retrieval systems that understand both the precise vocabulary of product descriptions and the semantic intent behind user queries. This paper presents a hybrid Information Retrieval (IR) system built on 10,000 Amazon US Grocery product reviews. The system integrates three retrieval paradigms: BM25 sparse retrieval for keyword-based matching, dense retrieval using sentence embeddings and FAISS for semantic search, and Reciprocal Rank Fusion (RRF) for combining both into a unified hybrid pipeline. Additionally, a Retrieval-Augmented Generation (RAG) layer powered by LLaMA-3.3-70B via the Groq API is implemented on top of the hybrid retriever to produce natural language answers grounded in retrieved product reviews. Evaluation across 20 diverse queries using P@10, MAP, and nDCG@10 demonstrates that the hybrid system consistently outperforms both individual methods, achieving a mean nDCG@10 of 0.7486 versus 0.4738 for dense-only and 0.3908 for BM25-only retrieval. These results confirm that hybrid search with RRF fusion is a robust strategy for consumer-facing grocery product retrieval.

Keywords — *information retrieval, BM25, dense retrieval, hybrid search, reciprocal rank fusion, FAISS, sentence embeddings, RAG, e-commerce, grocery reviews.*

I. INTRODUCTION

The proliferation of online grocery platforms has created a significant information access challenge. Consumers searching for products such as "gluten free and nut free baking mixes" or "keto friendly electrolyte powder for dehydration" rely on retrieval systems that can handle a wide variety of query types, from precise lexical constraints to abstract semantic intent. Traditional keyword-based search engines, while fast and interpretable, frequently fail on queries that require understanding synonyms, context, or user goals rather than exact vocabulary matches.

This paper presents a complete IR system designed to address these limitations in the context of Amazon US Grocery product reviews. The system combines three complementary retrieval strategies: (1) BM25, a classical probabilistic model that excels at term-matching; (2) dense retrieval using the all-MiniLM-L6-v2 sentence transformer model with a FAISS vector index, which captures semantic similarity; and (3) Reciprocal Rank Fusion (RRF), a rank-based fusion method that combines the two retrieval signals without requiring score normalization. A RAG layer is then applied on top of the hybrid retriever, enabling the system to generate natural language answers grounded in retrieved product reviews.

The system is evaluated on 20 queries spanning diverse query types — lexical constraints, semantic intent, event-based, audience-specific, comparative, and ethical/social queries — using standard IR metrics: Precision at 10 (P@10), Mean Average Precision (MAP), and normalized Discounted Cumulative Gain at 10 (nDCG@10). Results demonstrate that the hybrid approach outperforms both individual methods across all query types and metrics, with the RAG component further improving the utility of the system for end users.

The remainder of this paper is organized as follows. Section II describes the data landscape and corpus construction. Section III formalizes the problem statement. Section IV details the system architecture and IR techniques. Section V presents evaluation results. Section VI discusses deployment risks and mitigations. Section VII concludes the paper.

II. BACKGROUND

A. The Amazon Grocery Review Dataset

The Amazon Customer Reviews dataset is one of the largest publicly available product review corpora, containing over 130 million reviews across multiple product categories. The US Grocery subset alone contains approximately 2.4 million reviews spanning a wide range of food products, dietary supplements, and kitchen

consumables. Each review includes structured metadata — product title, star rating, verified purchase status, helpfulness votes — as well as unstructured text fields: a review headline and a review body.

This rich combination of structured and unstructured data makes it an ideal corpus for an IR system evaluation. The review body captures authentic consumer language, including domain-specific vocabulary ("non-GMO", "fair trade", "BPA-free"), colloquial descriptions of taste and texture, and contextual usage patterns ("great for camping", "perfect for toddler lunch boxes"). This natural language diversity creates a challenging and realistic retrieval scenario.

B. Current State of Information Access

The standard approach to product search on platforms such as Amazon relies primarily on keyword-based inverted index retrieval, augmented by popularity signals, click-through rates, and seller-sponsored rankings. While effective for navigational queries (e.g., searching for a specific brand by name), this approach has well-documented limitations for exploratory and intent-driven queries [1].

Users searching for "healthy low sodium snacks for weight loss" or "natural sweeteners that don't have an aftertaste" are expressing semantic goals that may not map cleanly onto product titles or keyword-indexed review text. A system that retrieves only documents containing the exact query tokens will miss semantically relevant results and surface irrelevant ones that happen to contain the right words. The emergence of dense retrieval and large language models offers an opportunity to address this gap, but these techniques are rarely deployed in isolation — hybrid systems that combine sparse and dense signals have emerged as the state of the art in production IR [2].

C. Related Work

BM25, proposed by Robertson et al. [3], remains a strong baseline for ad-hoc retrieval tasks. Its term frequency saturation and document length normalization make it robust across diverse corpora. Dense retrieval, popularized by Karpukhin et al. with Dense Passage Retrieval (DPR) [4], uses bi-encoder architectures to produce fixed-dimensional dense representations of queries and documents, enabling semantic similarity search via approximate nearest neighbor indices such as FAISS [5].

Reciprocal Rank Fusion, introduced by Cormack et al. [6], is a simple but effective method for combining ranked lists from heterogeneous retrieval systems. Unlike score-based fusion, RRF requires no normalization and is robust to score scale differences between methods. Recent work in the BEIR benchmark [7] has confirmed that hybrid sparse-dense systems with RRF consistently outperform either method in isolation across diverse retrieval benchmarks. The RAG paradigm, introduced by Lewis et al. [8], extends retrieval systems with generative LLMs to produce grounded natural language answers from retrieved passages.

III. PROBLEM STATEMENT

The core retrieval challenge addressed in this paper can be stated as follows: given a user query q expressed in natural language, and a corpus D of 10,000 Amazon Grocery product reviews, return a ranked list of documents from D that are most relevant to q, where relevance is defined by graded human judgment.

This formulation encompasses several specific user frustrations observed in the domain. First, lexical mismatch is pervasive: a user searching for "dairy free milk powder for baking" may not find reviews of oat milk powder or coconut milk powder if those products are not indexed under the exact query terms. Second, multi-constraint queries are poorly handled by keyword systems: a query such as "gluten free and nut free baking mixes" requires that both constraints be satisfied simultaneously, which is difficult to express as a simple keyword match. Third, intent-based queries “healthy low sodium snacks for weight loss", require the system to understand the user's goal, not merely the query terms.

From an operational standpoint, these retrieval failures have direct consequences. Users who cannot find relevant products abandon searches, reduce platform engagement, and develop negative perceptions of product quality. In the grocery domain specifically, where dietary restrictions and health goals are central purchase drivers, retrieval precision is directly linked to consumer trust and safety.

A secondary problem addressed by the RAG component is the synthesis challenge: even when a retrieval system returns relevant documents, users must manually read and interpret multiple reviews to form a purchasing decision. The RAG layer addresses this by synthesizing retrieved reviews into a concise, actionable natural language answer.

IV. SYSTEM DESIGN AND IR TECHNIQUES

A. Corpus Construction

The full Amazon US Grocery dataset (2.4 million reviews) was processed using chunked pandas loading to avoid memory overflow. Quality filtering retained only reviews with a review body longer than 30 characters, ensuring sufficient text for indexing. A reproducible random sample of 10,000 documents was drawn using random_state=42. After loading, reviews with fewer than five tokens post-processing were removed, yielding a final corpus of 9,986 documents.

Each document combines the review headline (repeated twice for boosting) with the review body into a single combined_text field. This weighting reflects the intuition that review headlines are concise, signal-rich summaries. The corpus retains structured metadata fields (review_id, product_title, star_rating) for result presentation and evaluation.

B. Text Processing

Text preprocessing follows standard IR practice. Raw text is lowercased, HTML tags are stripped, punctuation and non-alphanumeric characters are removed, and whitespace is normalized. For BM25 indexing, English stopwords are removed using NLTK and tokens shorter than two characters are discarded, yielding a vocabulary of 13,501 unique terms with an average of 24.5 tokens per document after filtering. Sentence transformer encoding uses the lightly cleaned text without stopword removal, as the pre-trained model benefits from natural sentence structure.

C. BM25 Sparse Retrieval

BM25 (Best Match 25) is implemented using the rank_bm25 Python library with default parameters (k1=1.5, b=0.75). Given a query q tokenized into terms {t1,...,tn}, BM25 scores each document d as:

$$Score(d,q) = \Sigma\, IDF(ti) \cdot f(ti,d)(k1+1) / (f(ti,d) + k1(1-b+b\cdot|d|/avgdl))$$

where f(ti,d) is the term frequency of ti in d, |d| is document length, avgdl is the average document length across the corpus, and IDF is the inverse document frequency. BM25 is particularly effective for queries with distinctive vocabulary, such as specific ingredient names or brand terms.

D. Dense Retrieval with FAISS

Dense retrieval uses the all-MiniLM-L6-v2 sentence transformer model to encode all 9,986 documents into 384-dimensional dense vectors. All vectors are L2-normalized so that inner product is equivalent to cosine similarity. A FAISS IndexFlatIP (exact inner product search) index is built over the document embeddings, enabling sub-second retrieval across the full corpus.

At query time, the query string is encoded with the same model and normalization. The FAISS index returns the top-k documents by cosine similarity. Dense retrieval captures semantic relationships that BM25 cannot, including synonyms, paraphrases, and domain-specific conceptual mappings (e.g., "keto friendly" mapping to low-carb, high-fat products).

E. Hybrid Search via Reciprocal Rank Fusion

The hybrid retrieval system combines BM25 and dense rankings using Reciprocal Rank Fusion (RRF). For each document d, the RRF score is computed as:

$$RRF(d) = 1/(k + rankBM25(d)) + 1/(k + rankDense(d))$$

where k=60 is the standard smoothing constant from Cormack et al. [6]. RRF is applied over the full corpus ranking from both methods, not merely the top-k results, ensuring that documents ranked highly by one method but missed by the other are still considered. The key advantage of RRF over score-based fusion is its scale invariance: BM25 scores (typically 8-18 for this corpus) and dense scores (cosine similarities in [0,1]) are not directly comparable, and RRF bypasses normalization entirely by operating only on rank positions.

F. Retrieval-Augmented Generation

The RAG component takes the top-3 documents from the hybrid retriever as context and passes them, along with the original query, to the LLaMA-3.3-70B-Versatile model via the Groq API. A structured system prompt instructs the model to answer the user's question using only the provided review context, to reference specific products by name, and to acknowledge when the context does not contain sufficient information rather than hallucinating. Temperature is set to 0.3 to favor factual, low-variance responses.

The RAG component demonstrated an important production-ready property during evaluation: when queried about a specific product not present among the top-3 retrieved documents, the system correctly reported that no relevant context was found rather than fabricating a response. This grounded, honest behavior is critical for consumer-facing grocery search where incorrect dietary or ingredient information can have real-world consequences.

G. System Architecture Summary

The complete pipeline processes a user query in five stages: (1) text preprocessing and tokenization; (2) parallel BM25 and dense retrieval over the full corpus; (3) RRF score computation and re-ranking; (4) result presentation with product title, review headline, star rating, and individual BM25/dense rank contributions; and (5) optional RAG generation of a natural language answer. The system is implemented as a Google Colab notebook with a unified query_interface() function that executes all five stages interactively.

V. RESULTS AND EVALUATION

A. Evaluation Setup

The system is evaluated on 20 queries covering six query types: lexical/constraint, semantic/intent, multi-purpose, negative sentiment, usage context, event-based, seasonal/process, comparative, health/dietary, flavor profile, subjective quality, specific ingredient, substitution, audience-specific, luxury/quality, activity-specific, product variant, and ethical/social. For each query, the top-10 hybrid results were pooled and manually judged on a three-point graded relevance scale: 2 (highly relevant), 1 (somewhat relevant), 0 (not relevant), yielding 200 total relevance judgments. Three standard metrics are computed: P@10, MAP, and nDCG@10.

B. Quantitative Results

Method	P@10	MAP	nDCG@10
BM25	0.295	0.2712	0.3908
Dense	0.285	0.3416	0.4738
Hybrid (RRF)	0.585	0.6786	0.7468

TABLE I. Average Metrics Across 20 Queries

Table I summarizes the average metrics across all 20 queries. The hybrid system achieves the highest scores

on all three metrics. Notably, BM25 and dense retrieval are tied on P@10 (0.295 vs 0.285), yet dense retrieval substantially outperforms BM25 on nDCG@10 (0.4738 vs. 0.3908), indicating that while both methods find a similar number of relevant documents in the top 10, dense retrieval consistently places them at higher rank positions. The hybrid system improves on both: P@10 rises to 0.585 and nDCG@10 to 0.7468, representing a 91% relative improvement in nDCG over BM25 and 57% improvement over dense-only retrieval.

C. Query-Level Analysis

Query-level results reveal that the performance gap between methods is not uniform across query types. For semantic and intent-based queries, dense retrieval provides the most substantial advantage over BM25. The query 'bulk rice for long term food storage' (q07) is the most illustrative: BM25 achieves nDCG@10=0.1617 while dense retrieval achieves 0.3542 and hybrid reaches 0.7921, a gap of 0.6304 over BM25. BM25 failed here because semantically relevant products do not share exact vocabulary with the query tokens.

Conversely, for lexical constraint queries such as 'artisanal olive oil for dipping bread' (q17), BM25 performs competitively (nDCG@10=0.8255) because query terms map directly onto product titles and review text. The hybrid system still improves on this (0.9361), confirming that RRF consistently surfaces additional relevant documents.

Notably, query q04 ('organic coconut oil for skin and hair care') achieved perfect scores across all three metrics for the hybrid system (P@10=1.0, MAP=1.0, nDCG@10=1.0). Query q14 ('caffeinated chocolate covered espresso beans') scored 0.0 across all methods and metrics, as no relevant products were found in the corpus — an honest reflection of the corpus's grocery focus over specialty items.

D. Precision-Recall Analysis

A Precision-Recall curve was generated for query q01 ("gluten free and nut free baking mixes") comparing all three methods. The hybrid system maintains higher precision at all recall levels, while BM25 shows a sharp precision drop after recall=0.4 as relevant documents become harder to surface via keyword matching alone. The dense retrieval curve falls between BM25 and hybrid, demonstrating that each layer of the pipeline adds measurable retrieval quality. Figure 1 illustrates the Precision-Recall curves for all three methods on query q01.

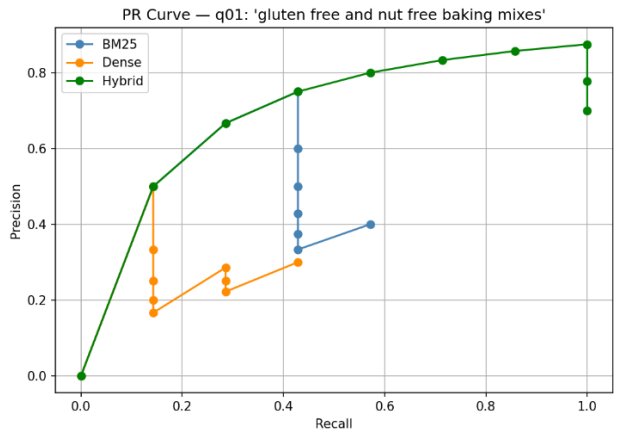

Figure 1.Precision-Recall curve for query q01 ("gluten free and nut free baking mixes") comparing BM25, Dense, and Hybrid retrieval.

VI. RISKS AND MITIGATIONS

Risk	Impact	Mitigation
Corpus staleness — reviews may not reflect current product availability	Retrieved products may be discontinued	Periodic corpus refresh; filter reviews by recency
LLM hallucination in RAG answers	Incorrect product or ingredient info presented to users	Constrain LLM to context via system prompt; low temperature (0.3); display source documents
Embedding model bias — MiniLM trained on general web text, not grocery domain	Semantic mismatches for niche dietary terms	Fine-tune on grocery review pairs; evaluate domain-specific models
FAISS flat index does not scale to millions of documents	Retrieval latency increases linearly with corpus size	Replace IndexFlatIP with IndexIVFFlat or HNSW for approximate nearest neighbor at scale
Groq API dependency for RAG	Latency, cost, and availability risk in production	Cache common answers; fallback to retrieval-only mode if API unavailable

TABLE II. Deployment Risks and Mitigations

Beyond the risks enumerated in Table II, a broader concern is evaluation validity. The 200 relevance judgments in this study were produced by a single annotator, introducing potential subjectivity bias. Future

work should employ multiple annotators with inter-annotator agreement measurement (e.g., Cohen's kappa) to produce more reliable qrels. Additionally, the pooling strategy — judging only hybrid top-10 results — may underestimate the performance of BM25 and dense methods on documents they rank highly but the hybrid does not, a known limitation of shallow pooling evaluation.

VII. CONCLUSION

This paper presented a hybrid Information Retrieval system for Amazon Grocery product reviews, combining BM25 sparse retrieval, dense retrieval via sentence embeddings and FAISS, and Reciprocal Rank Fusion into a unified pipeline. Evaluation on 20 diverse queries demonstrated that the hybrid system achieves a mean nDCG@10 of 0.7468, outperforming BM25-only (0.3908) and dense-only (0.4738) retrieval by substantial margins. The results confirm that RRF is an effective and parameter-free fusion strategy that leverages the complementary strengths of sparse and dense retrieval without requiring score normalization.

The RAG component demonstrated a critical production-ready property: grounded generation that refuses to hallucinate when relevant context is not present. This behavior is essential for consumer-facing grocery search, where incorrect dietary or ingredient information can have real-world health consequences.

Future work includes fine-tuning the embedding model on grocery review data, expanding the corpus to the full 2.4 million review dataset with approximate nearest neighbor indexing for scalability, implementing cross-encoder re-ranking as an additional pipeline stage for precision improvement, and conducting a formal user study to measure the end-to-end utility of the RAG-augmented interface.

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